

DM545/DM871 – Linear and integer programming

Exercise Sheet 4, Autumn 2026

Solution:

Included.

Exercises with the symbol $+$ are to be done at home before the class. Exercises with the symbol $*$ will be tackled in class and should be at least read at home. The remaining exercises are left for self training after the exercise class.

Exercise 1⁺

Show that the dual of $\max\{c^T x \mid Ax = b, x \geq 0\}$ is $\min\{y^T b \mid y^T A \geq c\}$ using the bounding method. Show that the dual of $\max\{c^T x \mid Ax \leq b, x \geq 0\}$ is $\min\{y^T b \mid y^T A \geq c, y \geq 0\}$ using the Lagrangian multipliers method.

Solution:

Given $\max\{c^T x \mid Ax = b, x \geq 0\}$ we search for multipliers $y \in \mathbb{R}^n$ such that $y^T Ax = y^T b$ (since we have equalities, the multipliers can be both positive or negative as we do not need to ensure the maintainance of the direction of the inequality). To ensure that we find an upper bound and hence have $c^T x \leq y^T Ax$, we impose $y^T A \geq c^T$ (since $x \geq 0$). Hence, the best upper bound will be given by solving $\min\{y^T b \mid y^T A \geq c^T\}$ (recalling from linear algebra that $(AB)^T = B^T A^T$, we can rewrite: $\min\{y^T b \mid A^T y \geq c\}$, which is the form we would obtain using the recipe method.)

$$\begin{array}{ll} \min & z = c^T x & c \in \mathbb{R}^n \\ & Ax \leq b & A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m \\ & x \geq 0 & x \in \mathbb{R}^n \end{array}$$

We want to create a relaxation by enlarging the feasibility region and ensuring that in the feasibility region the new objective function stays above $c^T x$ because we want to approximate the function from above. For feasible solutions to the original problem we have: $b - Ax \geq 0$. Hence, when we relax this quantity in the objective function we multiply by nonnegative coefficients $y \in \mathbb{R}_+^n$.

$$\begin{aligned} & \max_{y \in \mathbb{R}_+^m} \{ \min_{x \in \mathbb{R}_+^n} \{ c^T x + y^T (b - Ax) \} \} \\ & \max_{y \in \mathbb{R}_+^m} \{ \min_{x \in \mathbb{R}_+^n} \{ (c^T - y^T A)x + y^T b \} \} \end{aligned}$$

The inner problem is easy to solve when we are trying to maximize the outer problem since this latter one we will try to make $A^T y \leq c$. Hence, we are left with the problem

$$\begin{array}{ll} \max & b^T y \\ & A^T y \leq c \\ & y \in \mathbb{R}_+^m \end{array}$$

Exercise 2^{*}

Consider the following LP problem:

$$\begin{array}{ll} \max & 2x_1 + 3x_2 \\ & 2x_1 + 3x_2 \leq 30 \\ & x_1 + 2x_2 \geq 10 \\ & x_1 - x_2 \leq 1 \\ & x_2 - x_1 \leq 1 \\ & x_1 \geq 0 \end{array}$$

- Write the dual problem
- Using the optimality conditions derived from the theory of duality, and without using the simplex method, find the optimal solution of the dual knowing that the optimal solution of the primal is $(27/5, 32/5)$.

Solution:

The dual is:

$$\begin{aligned} \min \quad & 30y_1 + 10y_2 + y_3 + y_4 \\ & 2y_1 + y_2 + y_3 - y_4 \geq 2 \\ & 3y_1 + 2y_2 - y_3 + y_4 = 3 \\ & y_1, y_3, y_4 \geq 0 \\ & y_2 \leq 0 \end{aligned}$$

We use the complementary slackness theorem.

$$\begin{cases} 2y_1 + y_2 + y_3 - y_4 = 2 \\ 3y_1 + 2y_2 - y_3 + y_4 = 3 \\ y_2 = 0 \\ y_3 = 0 \end{cases}$$

The first because the corresponding variable of the primal is > 0 , the second for the same reason or however because it is already tight by definition, the third and fourth equation are a consequence of the fact that substituting the value of the primal variables variables in the primal problem, the second and third constraints are binding. What we obtain is a linear system of four equations in four variables that we can solve to find the value of the variables of the dual problem.

Exercise 3

Consider the problem

$$\begin{aligned} \text{maximize} \quad & 5x_1 + 4x_2 + 3x_3 \\ \text{subject to} \quad & 2x_1 + 3x_2 + x_3 \leq 5 \\ & 4x_1 + x_2 + 2x_3 \leq 11 \\ & 3x_1 + 4x_2 + 2x_3 \leq 8 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Without applying the simplex method, how can you tell whether the solution $(2, 0, 1)$ is an optimal solution? Is it? [Hint: consider consequences of Complementary slackness theorem.]

Solution:

The primal problem:

$$\begin{aligned} \max \quad & 5x_1 + 4x_2 + 3x_3 \\ \text{s.t} \quad & 2x_1 + 3x_2 + x_3 \leq 5 \\ & 4x_1 + x_2 + 2x_3 \leq 11 \\ & 3x_1 + 4x_2 + 2x_3 \leq 8 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

The dual problem:

$$\begin{aligned} \min \quad & 5y_1 + 11y_2 + 8y_3 \\ \text{s.t} \quad & 2y_1 + 4y_2 + 3y_3 \geq 5 \\ & 3y_1 + y_2 + 4y_3 \geq 4 \\ & y_1 + 2y_2 + 2y_3 \geq 3 \\ & y_1, y_2, y_3 \geq 0 \end{aligned}$$

If the Complementary slackness theorem holds and the objective value for the primal and dual is the same, then the solution $(2, 0, 1)$ is optimal. We check that it does:
Since $x_1, x_3 > 0$ we have the two constraints:

$$2y_1 + 4y_2 + 3y_3 = 5$$

$$y_1 + 2y_2 + 2y_3 = 3$$

Now we can test which constraints are binding in the primal problem.

We can see that the second constraint is non-binding, then $y_2 = 0$ and we have the following system

$$2y_1 + 4y_2 + 3y_3 = 5$$

of equations: $y_1 + 2y_2 + 2y_3 = 3$

$$y_2 = 0$$

Then we have that:

$$2y_1 + 3y_3 = 5$$

$$y_1 + 2y_3 = 3$$

Solved as before will give $y_1 = 1, y_3 = 1$.

Then the solution is:

$$y_1^* = 1, y_2^* = 0, y_3^* = 1$$

We need to check if it is feasible. We have already checked the conditions for the primal problem and we have already checked the first and third constraint in the dual, so we need to check the second constraint of the dual. It needs to be non-binding since $x_2 = 0$:

$$3y_1^* + y_2^* + 4y_3^* > 4$$

$$\Rightarrow 3 * 1 + 0 + 4 * 1 > 4$$

$$\Rightarrow 7 > 4$$

So it's non-binding.

We check that the objective values are the same:

For the primal: $5 * 2 + 4 * 0 + 3 * 1 = 13$

For the dual: $5 * 1 + 11 * 0 + 8 * 1 = 13$

Then the solution is optimal.

Exercise 4

Consider the following LP:

$$\begin{aligned} \min \quad & 3x_1 + 2x_2 - 4x_3 \\ \text{s.t.} \quad & 2x_1 + x_2 + x_3 \geq 3 \\ & x_1 + x_2 + 2x_3 \leq 5 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Find the optimal solution knowing that the solution of the dual problem is $(u_1, u_2) = (10/3, 11/3)$.

Solution:

Rewrite:

$$\begin{aligned} \max \quad & -3x_1 - 2x_2 + 4x_3 \\ \text{s.t.} \quad & -2x_1 - x_2 - x_3 \leq -3 \\ & x_1 + x_2 + 2x_3 \leq 5 \\ & x_i \geq 0 \quad \forall i \in \{1, 2, 3\} \end{aligned}$$

The dual problem:

$$\begin{aligned} \min \quad & -3u_1 + 5u_2 \\ \text{s.t.} \quad & -2u_1 + u_2 \geq -3 \\ & -u_1 + u_2 \geq -2 \\ & -u_1 + 2u_2 \geq 4 \\ & u_1, u_2 \geq 0 \end{aligned}$$

Now we want to find a solution to the primal. Since $u_1, u_2 > 0$ then we have the two constraints:

$$-2x_1 - x_2 - x_3 = -3$$

$$x_1 + x_2 + 2x_3 = 5$$

We check which constraints are binding in the dual:

We see that the second constraint is non-binding, hence $x_2 = 0$. And we have the following system of equations:

$$\begin{aligned} -2x_1 - x_2 - x_3 &= -3 \\ x_1 + x_2 + 2x_3 &= 5 \\ x_2 &= 0 \end{aligned}$$

Hence we have:

$$\begin{aligned} -2x_1 - x_3 &= -3 \\ x_1 + 2x_3 &= 5 \implies x_1 = 5 - 2x_3 \end{aligned}$$

Then we can insert x_1 in the first equation:

$$\implies -2(5 - 2x_3) - x_3 = -3 \implies -10 + 3x_3 = -3 \implies 3x_3 = 7 \implies x_3 = 7/3$$

And we can find x_1 :

$$x_1 = 5 - 2x_3 = 5 - 2 * 7/3 = 1/3$$

Then the solution of the primal is: $x_1 = 1/3, x_2 = 0, x_3 = 7/3$

Exercise 5*

Write the dual of the following problem

$$\begin{aligned} (P) \quad \max \quad & \sum_{j \in J} \sum_{i \in I} r_j x_{ij} \\ & \sum_{j \in J} x_{ij} \leq b_i & \forall i \in I \\ & \sum_{i \in I} x_{ij} \leq d_j & \forall j \in J \\ & \sum_{i \in I} p_i x_{ij} = p_j \sum_{i \in I} x_{ij} & \forall j \in J \\ & x_{ij} \geq 0 & \forall i \in I, j \in J \end{aligned}$$

Solution:

There are three different sets of constraints. We introduce the dual variables $\alpha_i \geq 0$, for $i \in I$, for the first set; the dual variables $\beta_j \geq 0$, for $j \in J$, for the second set; and the dual variables $\gamma_j \in \mathbb{R}$, for $j \in J$ for the third set.

We then write the A matrix for the example in the picture, augmented with the b vector: and finally the dual from the columns of the A matrix in general terms:

$$\begin{aligned} (D) \quad \min \quad & \sum_{i \in I} \alpha_i b_i + \sum_{j \in J} \beta_j d_j \\ & \alpha_i + \beta_j + (p_i - p_j)\gamma_j \geq r_j & \forall i \in I, j \in J \\ & \alpha_i \geq 0 & \forall i \in I \\ & \beta_j \geq 0 & \forall j \in J \\ & \gamma_j \in \mathbb{R} & \forall j \in J \end{aligned}$$

Exercise 6* Level Terrain for a New Highway

The Highway Department is planning a new 10-km highway on uneven terrain as shown by the profile in Figure 1. The width of the construction terrain is approximately 50 meters and the differences in height measured in meters can be deduced from the y -axis in the figure. To simplify the situation, the terrain profile can be replaced by a step function as shown in the figure. Using heavy machinery, earth removed from high terrain is pulled with effort to fill low areas. There are also two burrow pits, I and

$$\begin{array}{cccccccc}
 x_{G1} & x_{G2} & x_{G3} & x_{G4} & x_{T1} & x_{T2} & x_{T3} & x_{T4} \\
 \left[\begin{array}{cccccccc|c}
 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & b_1 \\
 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & b_2 \\
 1 & & & & 1 & & & & d_1 \\
 & 1 & & & & 1 & & & d_2 \\
 & & 1 & & & & 1 & & d_3 \\
 & & & 1 & & & & 1 & d_4 \\
 p_G - p1 & & & & p_T - p1 & & & & \\
 & p_G - p2 & & & & p_T - p2 & & & \\
 & & p_G - p3 & & & & p_T - p3 & & \\
 & & & p_G - p4 & & & & p_T - p4 &
 \end{array} \right]
 \end{array}$$

$r_1 \quad r_2 \quad r_3 \quad r_4 \quad r_1 \quad r_2 \quad r_3 \quad r_4$

II, located at the ends of the 10-km stretch from which additional earth can be pulled, if needed. Pit I is located before km 0 and has a capacity of 15 000 cubic meters, and pit II is located after km 10 and has a capacity of 11 000 cubic meters. The costs of removing earth from pits I and II are, respectively, 15 kr and 19 kr per cubic meter.

The transportation cost per cubic meter per kilometer is 1.5 kr, and the cost of using heavy machinery to load pulling trucks is 2 kr per cubic meter. This means that, for example, three cubic meters extracted from pit I and transported for 1 km will cost a total of $15 \cdot 3 + 1.5 \cdot 3 \cdot 1 + 2 \cdot 3 = 96$ kr and three cubic meters pulled 1 km away from a hill to a fill area will cost $1.5 \cdot 3 \cdot 1 + 2 \cdot 3 = 55.5$ kr.

Write a linear programming model for finding the minimum cost plan for leveling the 10-km stretch.

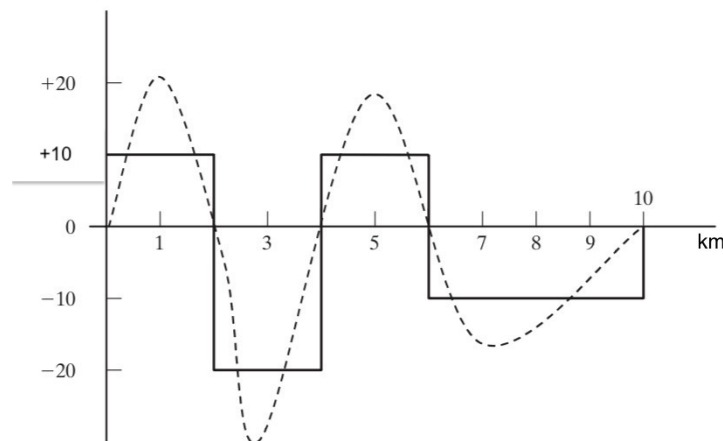


Figure 1:

Solution:

Let $i \in \{1..10\}$ denote the vertical parallelepiped with basis the segment of coordinates $i-1, i$ of length 1000 m. Let 0 indicate the pit I and 11 the pit II. Let a_i be the volume of the parallelepiped i . Let $I^+ \subset \{0..11\}$ be the set of parallelepiped that represent hill areas and $I^- \subset \{0..11\}$ be the set of those representing a fill area. The values of a_i can be calculated by calculating the volumes using the measures from the figure after proper transformation from kilometers to meters and by using the information about the width of 50 meters given in the problem. Let also $a_0 = 15000$ and $a_{11} = 11000$. Let d_{ij} be the distance between the centers of the parallelepiped i and j . The values of d_{ij} are expressed in meters and can be deduced from the figure.

Let $x_{ij} \geq 0, i, j = 1..10$ be the earth moved from parallelepiped i to position j .

$$\min \sum_{i=1}^{10} \sum_{j=1}^{10} (2 + 1.5 \cdot d_{ij}) x_{ij} + \sum_{j=1}^{10} (17 + 1.5 d_{0j}) x_{0j} + \sum_{j=1}^{10} (21 + 1.5 d_{11,j}) x_{11,j}$$

$$\sum_{i=0}^{11} x_{ij} = a_j \quad \forall j \in I^-$$

$$\sum_{j=1}^{10} x_{ij} \leq a_i \quad \forall i \in I^+$$

$$x_{ij} \geq 0 \quad \forall i, j = 1..10$$

The objective function measures the cost of moving earth around. The first component calculates the cost between the areas and the last two the cost of moving earth from the pit I and II, respectively. The first set of constraints ensure that the earth moved to the areas of low terrain is enough to fill the holes. The second set of constraints ensure that the earth moved away from the high areas does not make them become low areas. Finally, all earth movements are non negative.